MD-STGAT: A Multi-Modal Decoupled Spatio Temporal Graph Attention Network for Air Quality Forecasting

FINAL PROJECT REVIEW

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Problem Statement & Objectives

Background

- PM2.5 forecasting supports air-quality monitoring and early warning.
- Pollution evolves across both time and neighboring stations.
- Meteorological conditions influence local pollution dynamics.

Implementation Challenges

- Long gaps and missing wind-direction observations.
- Capturing 12-station spatial relationships.
- Combining pollution and weather without data leakage.
- Evaluating multi-horizon prediction accuracy.

Final Review Objectives

- Develop a 72-hour input → 24-hour PM2.5 forecasting pipeline for 12 Beijing stations.
- Implement separate pollution/weather encoders, spatial GATs, cross-attention, and temporal Transformer.
- Evaluate against a trained baseline; add decomposition and spatial regularization as subsequent research extensions.

Research Gap and Proposed Contribution

Research gap

Air-quality forecasting requires modeling temporal pollution patterns, spatial dependencies between monitoring stations, and meteorological influences. Our project investigates whether decoupled multimodal representations and attention-based fusion improve forecasting performance.

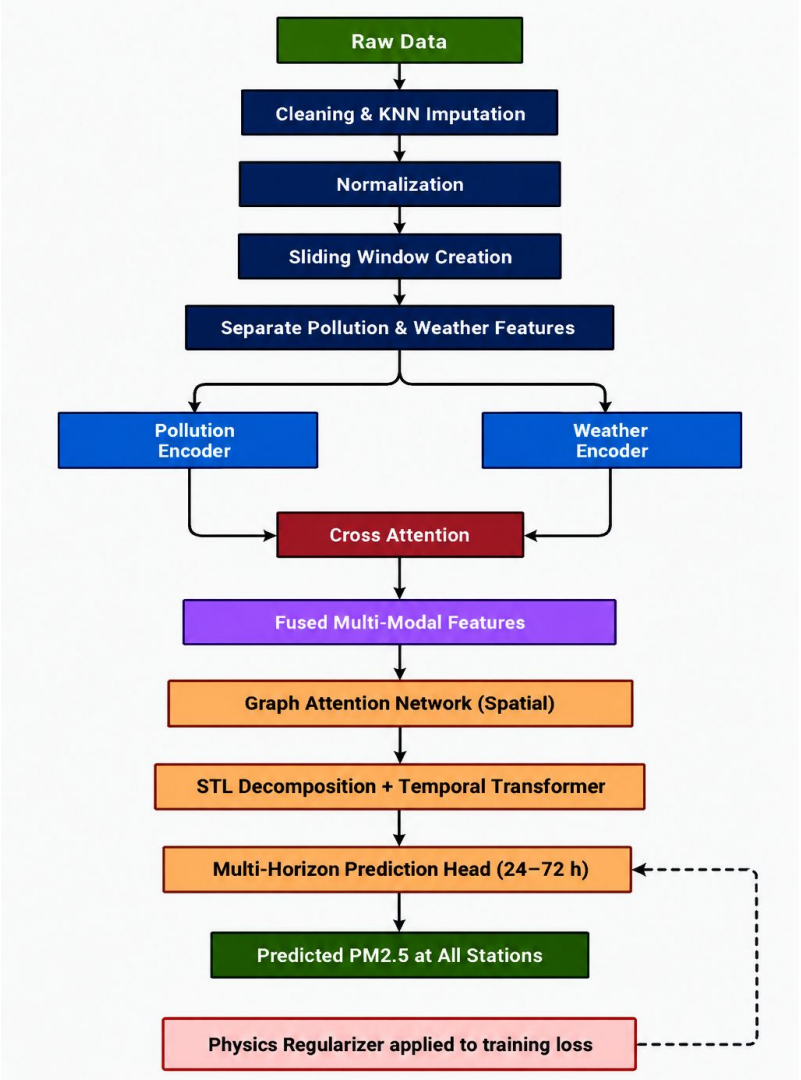
Proposed contributions

- Independent pollution and weather encoders.
- Separate spatial Graph Attention Networks.
- Weather-conditioned cross-modal attention.
- Temporal Transformer for historical sequence modeling.
- Direct 24-hour PM2.5 forecasting across 12 stations.

MD-STGAT Framework

Current implementation

- Data: chronological split, imputation, scaling, 72→24 h windows.
- Separate pollution and weather encoders with independent spatial GATs.
- Weather-query / pollution-key-value cross-attention; temporal Transformer.
- Full model forward pass verified on CUDA: output [B,24,12].
- Diagram also shows planned decomposition and physics-guided loss; not yet integrated.



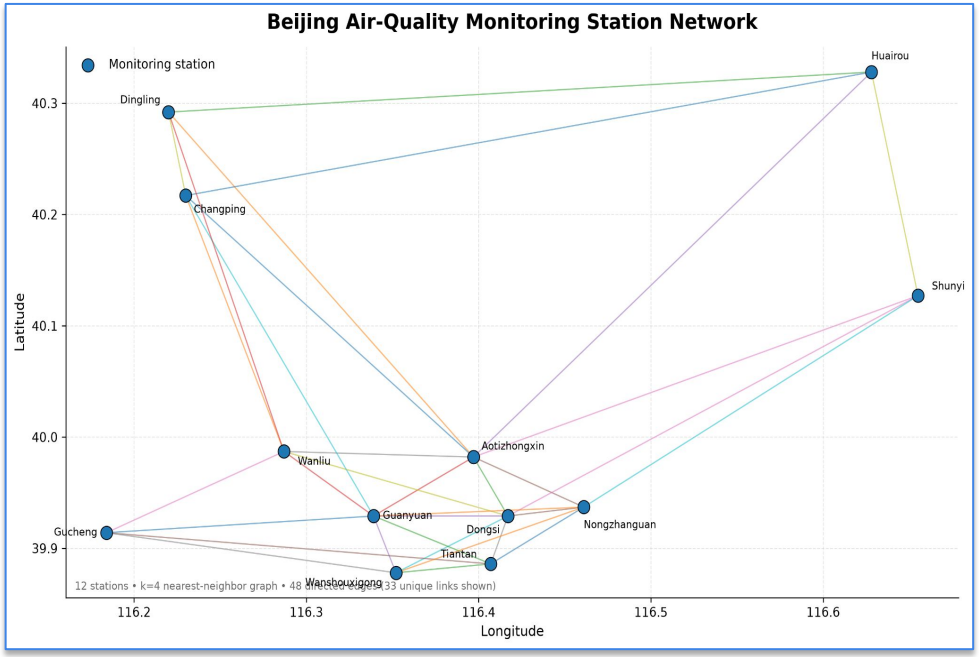
Dataset & Preprocessing Implementation

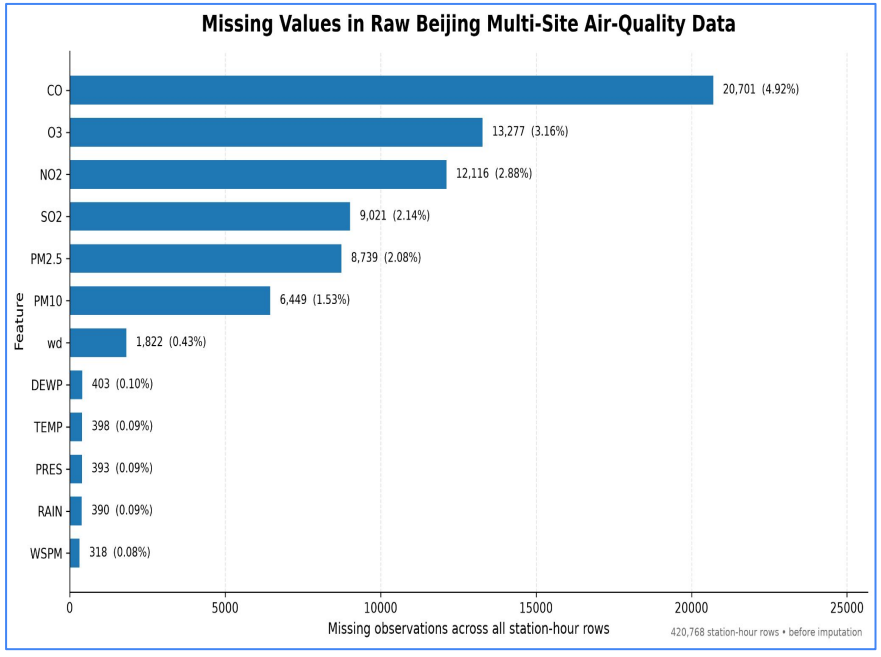
Dataset & Inputs

- UCI Beijing Multi-Site Air Quality dataset; 12 monitoring stations.
- 6 pollution features and 7 weather features.
- 72 historical hours → 24-hour PM2.5 forecast at all stations.

Completed Pipeline

- Chronological 70% / 15% / 15% split.
- Casual short-gap fill, spatial imputation, train-only scaling.
- 4-nearest-neighbor graph: 48 directed edges.





Implementation Status & Model Verification

Verified Components

- Pollution encoder + weather encoder.
- Independent spatial GAT per modality.
- Temporal cross-attention and Transformer.
- Complete forward pass on NVIDIA GPU.
- 170,456 trainable parameters.

Verified Tensor Shapes

- Pollution: [2,72,12,6].
- Weather: [2,72,12,7].
- Graph: [2,48].
- Forecast: [2,24,12].
- Attention: [2,12,72,72].

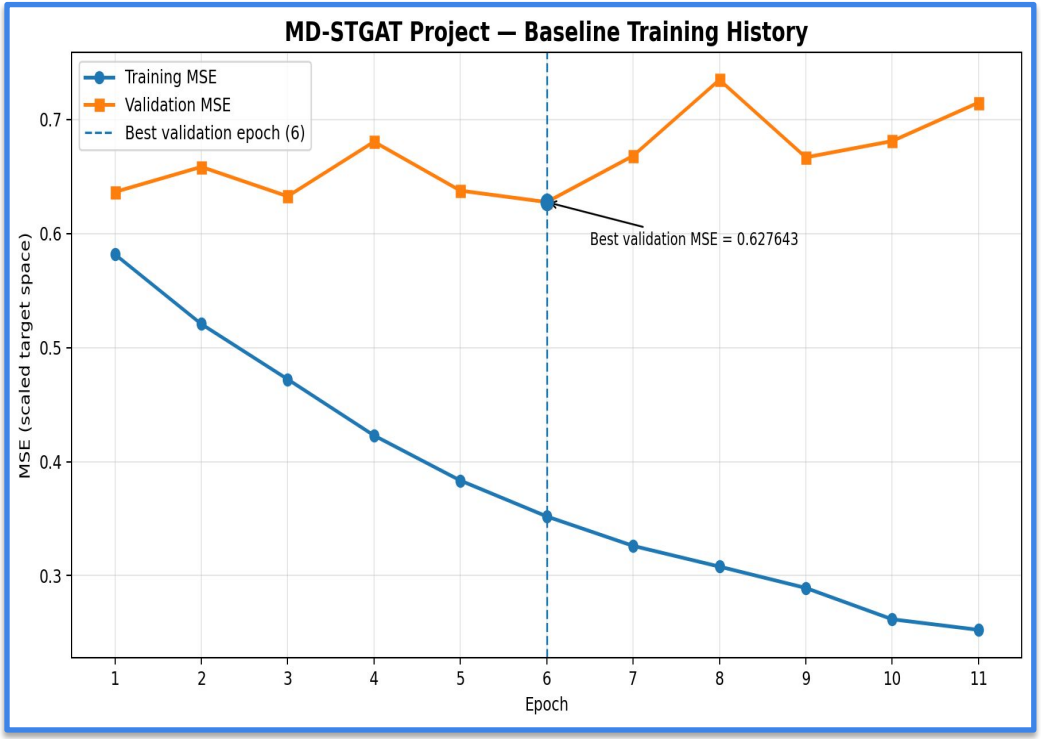
```
(dl_proj) ayush@ayush3112:~/dl_proj$ python -m models.md_stgat
/home/ayush/dl_proj/dl_proj/lib/python3.12/site-packages/torch/
 `torch.export`.
  warnings.warn(
Device: cuda
Original graph shape: (48, 2)
Graph shape used by GAT: (2, 48)
/home/ayush/dl_proj/models/temporal.py:51: UserWarning: enable_
  self.transformer = nn.TransformerEncoder(
Pollution input: (2, 72, 12, 6)
Weather input: (2, 72, 12, 7)
Forecast: (2, 24, 12)
Attention: (2, 12, 72, 72)
Trainable parameters: 170456
(dl_proj) ayush@ayush3112:~/dl_proj$
```

```
(dl_proj) ayush@ayush3112:~/dl_proj$ python -m models.cross_attention
Pollution: torch.Size([4, 72, 12, 64])
Weather: torch.Size([4, 72, 12, 64])
Output: torch.Size([4, 72, 12, 64])
Attention: torch.Size([4, 12, 72, 72])
Parameters: 49984
(dl_proj) ayush@ayush3112:~/dl_proj$
```

Baseline Training & Evaluation

Trained Baseline

- Pollution Encoder → Spatial GAT → Temporal Transformer → Forecast Head.
- Best validation MSE: 0.627643 (epoch 6).
- Test MAE: 45.1195 μg/m³.
- Test RMSE: 74.3686 μg/m³.
- Test R²: 0.3599.



Full Model Results & Remaining Work

Current Status

- Full MD-STGAT forward pass succeeds.
- Training blocked by non-finite wd_sin / wd_cos values.
- No validated full-model test metrics yet.

Next Implementation Steps

- Repair wind-direction imputation and regenerate datasets.
- Train/evaluate MD-STGAT; compare against baseline.
- Test temporal decomposition and graph smoothness loss.

What the MD-STGAT Test Results Tell Us

LOWEST STATION ERROR

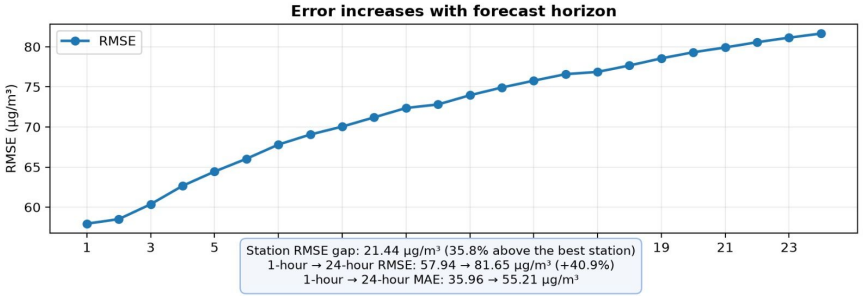
Dingling

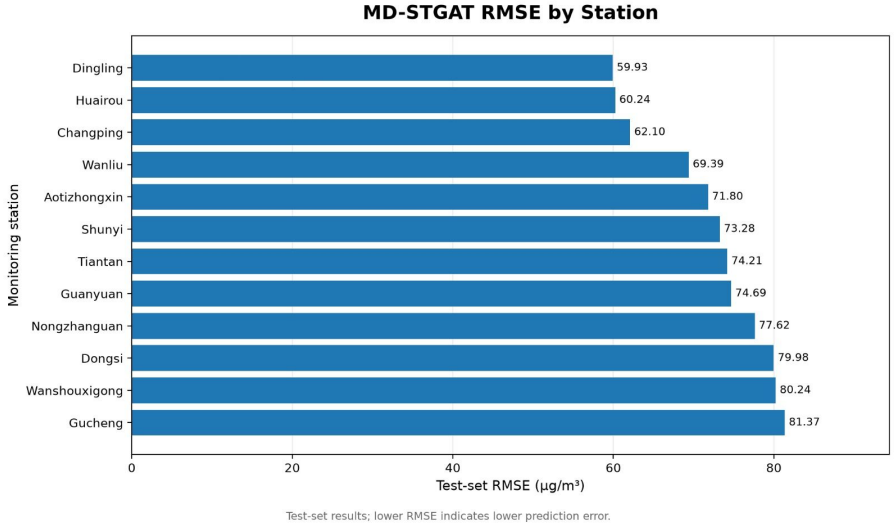
RMSE: 59.93 μg/m³

HIGHEST STATION ERROR

Gucheng

RMSE: 81.37 μg/m³





References

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